1 A student is investigating the characteristics of different light-emitting diodes (LEDs). Fig. 1.1 shows examples of LEDs and the circuit symbol for an LED.

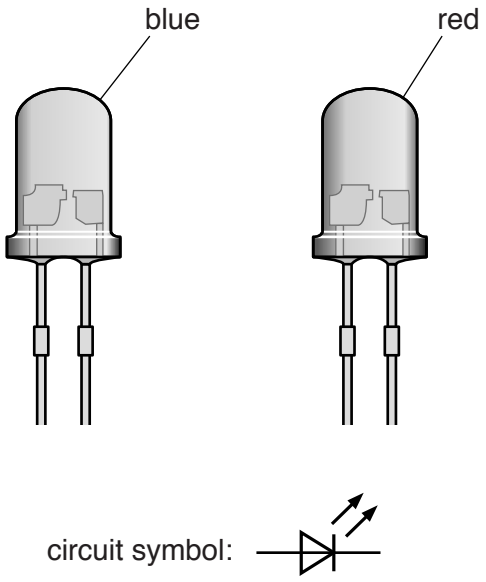

Fig 1.1

Each LED needs a minimum potential difference V across it to emit light. The student is investigating the relationship between V and the wavelength λ of the light emitted by the LED for several different LEDs.

It is suggested that the relationship is

$$V = k\lambda^n$$

where k and n are constants.

Design a laboratory experiment to test the relationship between V and λ. Explain how your results could be used to determine values for k and n. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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